



Course code: MEE 503L

Course Title: Finite Element Analysis Lab

Project Title: Structural Evaluation of Bicycle Frame Materials Under High Load

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1- Introduction

Bicycle frames are critical structural components that must safely support rider weight and operational loads without failure. With increasing rider mass, cargo usage, and demand for durability, the selection of an appropriate frame material becomes a key engineering challenge. Frame failure can lead to serious safety risks; therefore, understanding how different materials respond to high loading conditions is essential.

Finite Element Analysis (FEA) provides an effective numerical approach to evaluate stress distribution, deformation, and failure risk in complex structures such as bicycle frames. In this project, Abaqus is used to simulate a bicycle frame subjected to high vertical loading, and several commonly used frame materials are compared to determine which material offers the highest resistance to failure.

2- Problem Description

The purpose of this project is to evaluate the structural performance of a bicycle frame subjected to high loading conditions and to determine the most suitable material that prevents structural failure. The analysis focuses on comparing different frame materials under identical geometric, boundary, and loading conditions using finite element analysis in Abaqus.

A static structural approach is adopted, and failure is assessed based on the Von Mises stress criterion. The same bicycle frame model is analyzed for each material to ensure consistency and fairness in the comparison.

2-1 Design Load Definition

The design load applied to the bicycle frame represents the combined weight of the rider and any additional cargo. For this study, the total mass is assumed to be within the range typically considered high for bicycle usage.

Assumptions:

- Rider + cargo mass: 150–180 kg

The gravitational force acting on the frame is calculated using:

$$F = mg$$
$$F = 180 \times 9.81 \approx 1765 \text{ N}$$

To account for dynamic effects such as road irregularities, pedaling impacts, and sudden load variations, a dynamic amplification factor of 2 is applied, as commonly recommended in engineering design studies.

$$F_{design} = 2 \times 1765 \approx 3500 \text{ N}$$

The resulting design load of 3500 N is applied to the bicycle frame in the simulation to represent a conservative loading condition. This approach ensures that the evaluated materials are assessed under realistic and safety-critical conditions.

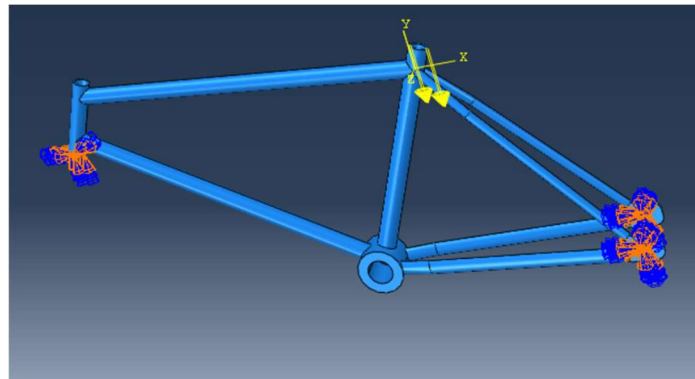


Figure 1-Applied B.C and loadings on Bicycle frame

3- Result

This section presents the finite element analysis results for three selected bicycle frame materials: Aluminum Alloy 7005, Titanium Alloy (Ti-6Al-4V), and High-Strength Steel (AISI 4130). For each material, the mechanical properties used in the simulation are summarized, followed by the displacement and Von Mises stress results obtained from Abaqus.

3-1 Aluminum Alloy 7005

Property	Value
Young's Modulus (GPa)	72
Poisson's Ratio	0.33
Yield Strength (MPa)	350

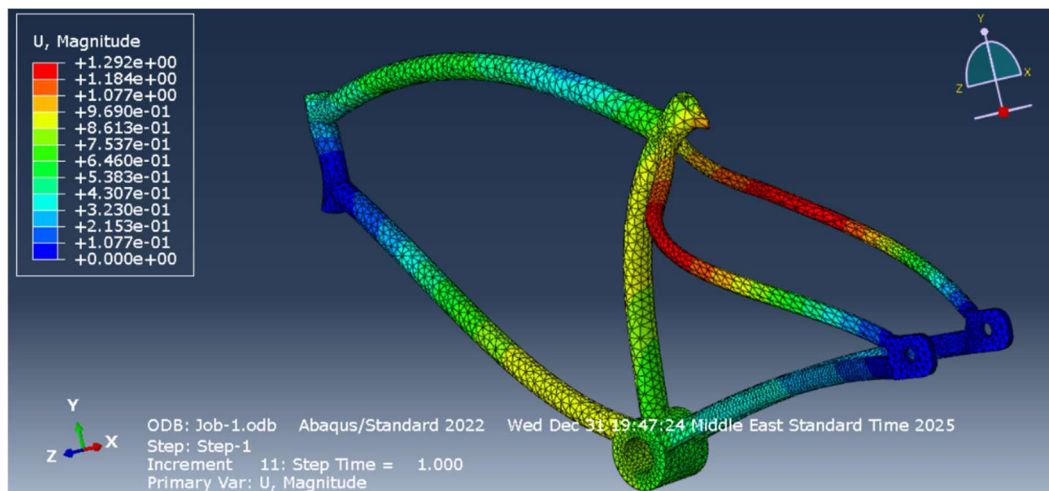


Figure 2-Displacement(Aluminum Alloy 7005)

Figure 2 shows the displacement contour plot of the bicycle frame made of Aluminum Alloy 7005 under the applied design load of 3500 N. The maximum displacement occurs primarily at the seat tube and top tube regions, indicating relatively low stiffness compared to other materials.

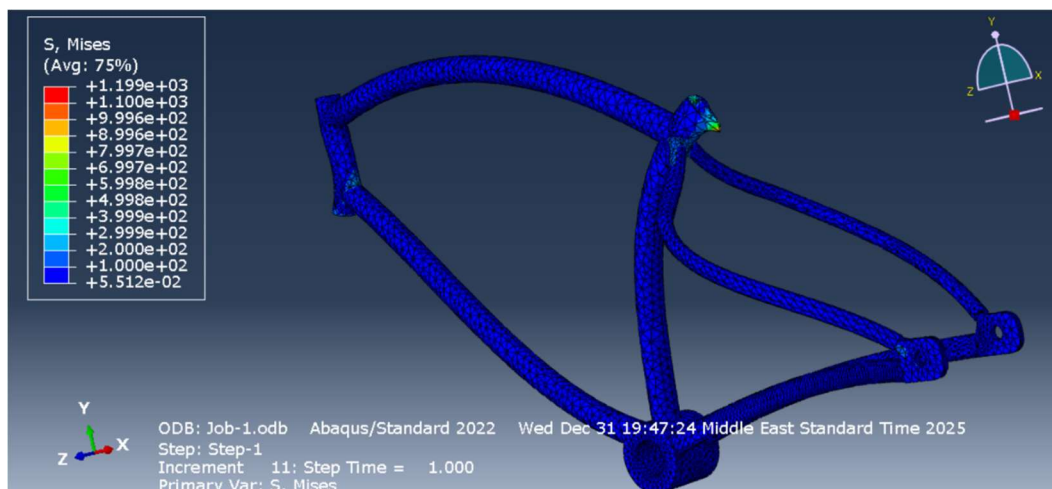


Figure 3-Maximum VON MISES(Aluminum Alloy 7005)

Figure 3 presents the Von Mises stress distribution for Aluminum Alloy 7005. The highest stress concentrations are observed near the bottom bracket and head tube junctions. These regions are critical and approach the yield limit of the material under high loading.

3-2 Titanium Alloy (Ti-6Al-4V)

Property	Value
Young's Modulus (GPa)	110
Poisson's Ratio	0.34
Yield Strength (MPa)	830

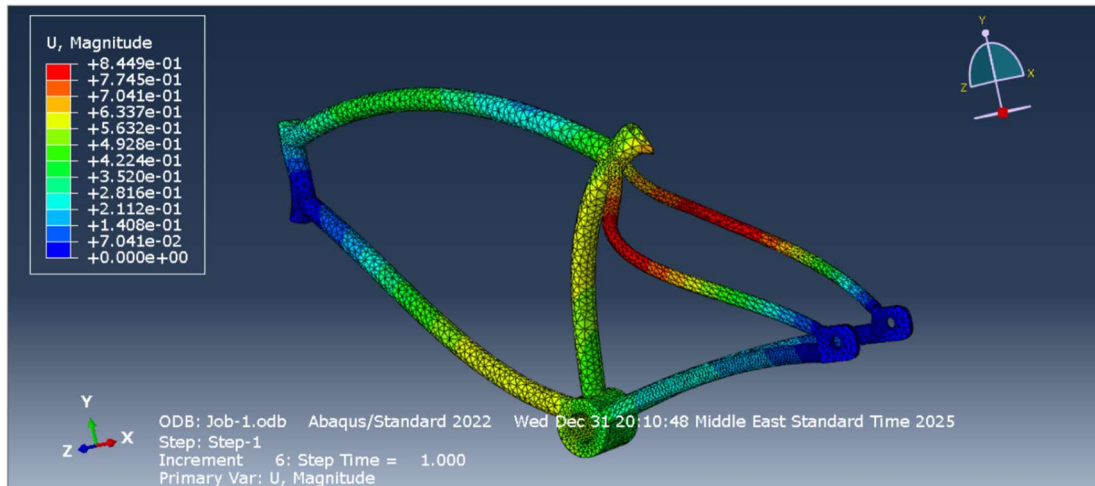


Figure 4-Displacement(Titanium Alloy (Ti-6Al-4V))

Figure 4 illustrates the displacement distribution of the bicycle frame constructed from Ti-6Al-4V. The overall deformation is lower than that observed in aluminum, reflecting the higher stiffness of titanium.

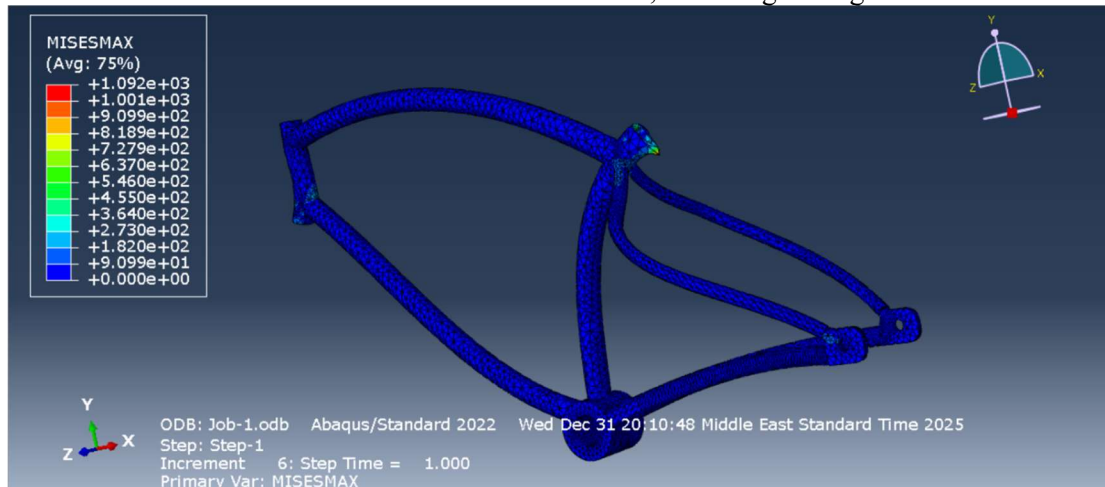


Figure 5-Maximum VON MISES(Titanium Alloy (Ti-6Al-4V))

Figure 5 shows the Von Mises stress contour for the titanium frame. Although stress levels are higher than those in steel due to lower stiffness, the stresses remain well below the yield strength, indicating a high safety margin.

3-3 High-Strength Steel (AISI 4130)

Property	Value
Young's Modulus (GPa)	210
Poisson's Ratio	0.29
Yield Strength (MPa)	435

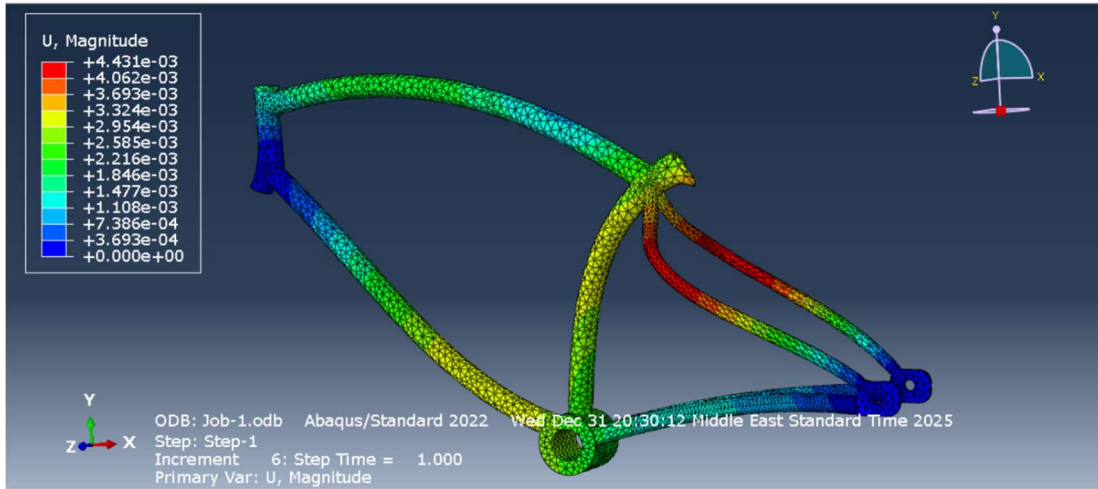


Figure 6-Displacement(High-Strength Steel (AISI 4130))

Figure 6 presents the displacement contour of the steel bicycle frame. The steel frame exhibits the lowest maximum displacement among the analyzed materials due to its high stiffness.

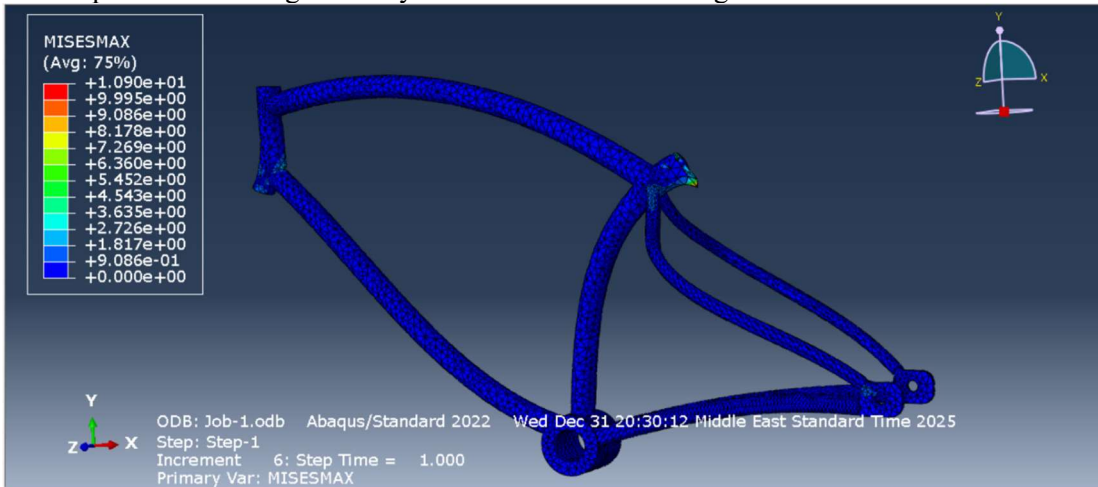


Figure 7-Maximum VON MISES(High-Strength Steel (AISI 4130))

Figure 7 displays the Von Mises stress distribution for AISI 4130 steel. Stress concentrations are observed at the same critical joints; however, the magnitude of stress remains significantly below the yield strength, demonstrating superior load-bearing capacity.

4- Discussion

In this section, the structural performance of each bicycle frame material is evaluated using the factor of safety (FoS) based on the Von Mises failure criterion. The factor of safety is calculated as:

$$FoS = \frac{\sigma_y}{\sigma_{vm,max}}$$

where:

- σ_y is the yield strength of the material
- $\sigma_{vm,max}$ is the maximum Von Mises stress obtained from Abaqus

A factor of safety greater than 1 indicates that the structure remains elastic, while values below 1 indicate yielding or failure.

4-1 Aluminum Alloy 7005

Yield strength: 350 MPa

Maximum Von Mises stress: 1199 MPa

$$FoS_{Al\ 7005} = \frac{350}{1199} \approx 0.29$$

The calculated factor of safety for Aluminum Alloy 7005 is significantly less than 1, indicating that the frame would experience severe yielding and structural failure under the applied design load. The high stress levels are primarily attributed to the relatively low stiffness and yield strength of aluminum compared to the applied load magnitude.

This result demonstrates that Aluminum Alloy 7005 is not suitable for high rider weight applications without substantial geometric reinforcement or increased tube thickness. The material fails to provide an adequate safety margin under the conservative loading conditions considered in this study.

4-2 Titanium Alloy (Ti-6Al-4V)

Yield strength: 830 MPa

Maximum Von Mises stress: 1092 MPa

$$FoS_{Ti-6Al-4V} = \frac{830}{1092} \approx 0.76$$

The titanium alloy frame exhibits a factor of safety below unity, indicating that yielding is expected under the applied design load. Although titanium has a very high yield strength and excellent fatigue resistance, the stresses developed in the frame exceed the elastic limit.

This suggests that, while Ti-6Al-4V performs substantially better than aluminum, it still does not meet the safety requirements for the given high-load condition in the current frame configuration. Design optimization, such as increasing tube diameters or wall thickness, would be necessary for titanium to be viable in this application.

4-3 High-Strength Steel (AISI 4130)

Yield strength: 435–560 MPa

Maximum Von Mises stress: 10.9 MPa

$$FoS_{steel\ AISI\ 4130} = \frac{435}{10.9} \approx 39.9$$

The high-strength steel frame exhibits an exceptionally large factor of safety, indicating that the stresses induced by the applied load are far below the material's yield limit. This behavior is a direct result of steel's high Young's modulus, which significantly reduces deformation and stress concentration.

The steel frame demonstrates outstanding structural reliability under high loading, making it the most suitable material among those studied for preventing failure. While the safety factor is much higher than typically required in engineering design, it provides confidence in durability and resistance to unexpected overloads.

5- Conclusion

This study investigated the structural behavior of a bicycle frame subjected to high loading conditions using finite element analysis in Abaqus. A conservative design load of 3500 N, representing a heavy rider and dynamic effects, was applied uniformly across all material cases. Aluminum Alloy 7005, Titanium Alloy (Ti-6Al-4V), and High-Strength Steel (AISI 4130) were evaluated using identical geometry, boundary conditions, and loading scenarios to ensure a fair comparison.

The results demonstrated a strong influence of material mechanical properties on frame performance. Aluminum Alloy 7005 exhibited excessive Von Mises stress exceeding its yield strength, resulting in a factor of safety well below unity and indicating structural failure under the applied load. Titanium Alloy (Ti-6Al-4V) showed improved performance compared to aluminum; however, the maximum stress still exceeded the yield limit, leading to an insufficient safety factor for high-weight applications without further design optimization.

In contrast, High-Strength Steel (AISI 4130) exhibited very low stress levels and minimal deformation, resulting in an exceptionally high factor of safety. This behavior is primarily attributed to steel's high stiffness and adequate yield strength, which effectively limit stress concentrations in critical regions such as the bottom bracket and head tube. The steel frame therefore demonstrated the highest resistance to failure and the greatest structural reliability among the materials studied.

Based on the finite element results and safety factor evaluation, it can be concluded that High-Strength Steel (AISI 4130) is the most suitable material for bicycle frames designed to carry high rider weight, providing superior safety and durability. While titanium offers advantages in terms of weight reduction, it requires geometric reinforcement to meet safety requirements. Aluminum alloys, although lightweight, are not suitable for high-load conditions in the current frame configuration.

Future work may include fatigue analysis, nonlinear material behavior, weld modeling, and geometry optimization to improve performance while reducing weight, as well as validation against international bicycle frame standards.